

HOPEDALE, NL, Canada

WMO: 719000

Lat: 55.449N

Lon: 60.228W

Elev: 12

StdP: 101.18

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.8	-26.0	-33.4	0.2	-26.9	-31.3	0.2	-25.1	19.0	-13.5	17.4	-14.6	6.4	270	0.900

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	22.2	15.6	19.9	14.6	17.9	13.6	16.3	20.4	15.2	19.0	14.1	17.4	6.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.4	10.2	18.2	13.2	9.5	17.4	12.1	8.8	16.3	45.9	20.7	42.7	19.0	39.8	17.5	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	13.7	12.1	DB	-30.1	27.6	2.0	1.7	-31.6	28.8	-32.8	29.8	-33.9	30.8	-35.4	32.0	(4)
(5)			WB	-29.8	19.0	2.0	1.2	-31.3	19.8	-32.5	20.5	-33.6	21.2	-35.1	22.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-15.7	-16.3	-10.8	-4.0	1.7	6.7	11.4	12.3	8.6	3.5	-2.7	-9.7
	DBStd	10.84	6.87	5.88	6.39	4.86	3.18	4.40	4.13	3.61	3.38	3.00	3.80	5.97
	HDD10.0	4289	796	736	646	419	258	122	34	17	67	203	381	611
	HDD18.3	7123	1054	969	904	669	515	351	218	190	292	460	631	869
	CDD10.0	214	0	0	0	0	1	22	77	88	25	1	0	0
	CDD18.3	6	0	0	0	0	0	1	2	3	0	0	0	0
	CDH23.3	42	0	0	0	0	0	8	12	21	2	0	0	0
	CDH26.7	6	0	0	0	0	0	1	1	4	0	0	0	0
Wind	WSAvg	5.5	6.6	6.2	6.0	5.4	4.6	4.6	4.3	4.4	5.4	5.8	6.6	6.8
Precipitation	PrecAvg	884	84	85	84	61	56	67	88	72	61	72	71	76
	PrecMax	1157	161	195	217	186	127	170	198	193	133	136	125	166
	PrecMin	519	20	6	13	11	17	14	29	18	16	27	16	25
	PrecStd	161	40	53	50	37	26	42	48	42	28	30	31	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	1.0	4.6	8.2	15.8	24.3	24.9	26.1	21.4	14.2	7.8	3.1
		MCWB	1.2	0.3	1.2	4.3	9.8	16.2	16.2	17.9	14.0	10.5	5.8	2.1
	2%	DB	0.1	-1.5	1.8	5.8	10.7	19.7	21.8	22.1	18.5	11.2	4.8	1.5
		MCWB	-0.3	-1.8	0.3	3.0	6.3	13.6	15.6	16.0	13.2	8.0	3.0	1.0
	5%	DB	-1.5	-4.0	0.6	4.1	8.1	16.3	19.5	19.7	16.1	9.3	3.2	0.3
		MCWB	-1.9	-4.5	-0.3	2.0	5.0	11.7	14.7	14.9	12.1	7.0	1.9	-0.1
	10%	DB	-4.1	-7.1	-0.9	2.3	6.1	13.2	17.5	17.6	13.9	7.5	2.1	-1.1
		MCWB	-4.4	-7.6	-1.7	0.8	3.6	9.5	13.6	13.8	10.3	5.3	1.1	-1.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.3	0.4	1.4	4.7	9.8	17.0	17.6	18.1	15.7	11.2	5.1	2.4
		MCDB	1.5	0.9	2.3	7.7	14.6	23.3	21.9	22.6	18.6	13.4	6.2	2.9
	2%	WB	-0.3	-2.0	0.4	3.4	6.7	14.3	16.2	16.4	13.9	9.0	2.8	1.1
		MCDB	0.0	-1.9	1.0	5.8	9.5	19.1	20.5	20.2	17.4	11.3	3.7	1.5
	5%	WB	-1.8	-5.1	-0.4	2.4	5.0	11.9	15.2	15.2	12.4	7.3	1.9	0.1
		MCDB	-1.5	-4.7	0.2	4.2	7.9	15.9	19.2	18.6	16.1	9.3	2.7	0.4
	10%	WB	-4.2	-7.6	-2.0	1.2	3.7	9.7	14.1	14.1	10.8	5.7	1.1	-1.5
		MCDB	-3.8	-7.0	-1.2	2.4	6.0	12.9	17.4	17.1	13.9	7.4	1.7	-1.1
Mean Daily Temperature Range	5% DB	MDBR	6.8	7.5	7.8	6.8	5.9	6.5	7.0	6.6	5.9	4.1	4.2	5.4
		MCDBR	7.8	10.5	8.1	8.9	10.4	12.7	11.0	10.9	10.0	7.3	5.4	5.2
		MCWBR	6.6	10.7	6.4	7.3	6.9	7.6	6.2	5.7	6.2	5.7	5.2	5.1
	5% WB	MCDBR	6.7	10.4	6.6	8.5	9.8	12.3	10.3	9.5	9.7	7.3	4.8	5.0
		MCWBR	6.7	10.4	6.1	6.9	6.9	7.9	6.4	5.9	6.7	5.8	4.6	5.1
Clear-Sky Solar Irradiance	taub	0.239	0.265	0.277	0.338	0.375	0.377	0.400	0.370	0.344	0.306	0.259	0.243	
	taud	2.228	2.213	2.204	2.071	2.156	2.338	2.358	2.443	2.471	2.525	2.357	2.162	
	Ebn at Noon	706	817	897	876	863	867	838	845	820	777	683	591	
	Edh at Noon	54	81	105	140	138	117	113	97	83	62	48	46	
All-Sky Solar Radiation	RadAvg	0.74	1.67	3.12	4.64	5.41	5.08	4.57	3.85	2.87	1.50	0.75	0.50	
	RadStd	0.08	0.21	0.39	0.34	0.56	0.49	0.39	0.32	0.23	0.10	0.07	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page